

Supplementary Material: Zero-Incoherence Capacity of Interactive Encoding Systems

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1 Full Lean Handle Ledger

This supplement provides the complete Lean handle ledger cited by the manuscript. It includes every handle identifier, declaration name, and source module path.

ID	Lean Handle / Source	ID	Lean Handle / Source
AFM1	FactMatroid.determinesFact_iff_mem_factSpan Ssot/FactMatroid.lean	AFM2	FactMatroid.factMatroid_indep_iff Ssot/FactMatroid.lean
AFM3	FactMatroid.basisFacts_card_eq_finrank Ssot/FactMatroid.lean	AFM4	FactMatroid.basisFacts_minimal Ssot/FactMatroid.lean
AFM5	FactMatroid.minimalDetermining_basis Ssot/FactMatroid.lean	AFM6	FactMatroid.factRankFinset_le_card Ssot/FactMatroid.lean
AFM7	FactMatroid.coordProjection_range_le_card Ssot/FactMatroid.lean	AFM8	FactMatroid.factRankFinset_eq_card_of_indepFacts Ssot/FactMatroid.lean
AFM9	FactMatroid.factRankFinset_eq_total_of_determinesAllFinset Ssot/FactMatroid.lean	AFM10	FactMatroid.factSpanFinset_eq_range_coordProjection_dualMap Ssot/FactMatroid.lean
AFM11	FactMatroid.factRankFinset_eq_finrank_range_coordProjection Ssot/FactMatroid.lean	AFM12	FactMatroid.factRankFinset_le_finrank_directionSpace Ssot/FactMatroid.lean
AFM13	FactMatroid.coordProjection_range_eq_card_of_indepFacts Ssot/FactMatroid.lean	AFM14	FactMatroid.coordProjection_range_eq_total_of_determinesAllFinset Ssot/FactMatroid.lean
AFM15	FactMatroid.finrank_productFamily_directions Ssot/FactMatroid.lean	AFM16	FactMatroid.range_coordProjection_productDirections_disjSum Ssot/FactMatroid.lean
AFM17	FactMatroid.factRankFinset_product_disjSum Ssot/FactMatroid.lean	AFM18	FactMatroid.indepFacts_productFamily_iff Ssot/FactMatroid.lean
AFM19	FactMatroid.factMatroid_product_indep_iff Ssot/FactMatroid.lean	BND1	ssot_upper_bound Ssot/Bounds.lean
BND2	non_ssot_lower_bound Ssot/Bounds.lean	BND3	ssot_advantage_unbounded Ssot/Bounds.lean
CAP2	ssot_guarantees_coherence Ssot/Coherence.lean	CAP3	non_ssot_permits_incoherence Ssot/Coherence.lean
CIA3	ClaimClosure.finite_counting_converse Ssot/ClaimClosure.lean	CIA4	ClaimClosure.info_dof_counting_contradiction Ssot/ClaimClosure.lean
COH1	dof_one_implies_coherent Ssot/Coherence.lean	COH2	dof_gt_one_incoherence_possible Ssot/Coherence.lean
CRG1	Deps.cargo_ssot_incomplete Ssot/LangDeps.lean	CRG2	Deps.cargo_has_introspection Ssot/LangDeps.lean
CRG3	Deps.cargo_no_hooks Ssot/LangDeps.lean	DBE1	Database.external_etl_ssot_incomplete Ssot/LangDatabase.lean
DBE2	Database.external_etl_no_hooks Ssot/LangDatabase.lean	DBE3	Database.external_etl_no_introspection Ssot/LangDatabase.lean

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ID	Lean Handle / Source	ID	Lean Handle / Source
DBV1	Database.engine_view_ssot_complete Ssot/LangDatabase.lean	DBV2	Database.engine_view_has_hooks Ssot/LangDatabase.lean
DBV3	Database.engine_view_has_introspection Ssot/LangDatabase.lean	DER2	ClaimClosure.derivation_preserves_coherence_core Ssot/ClaimClosure.lean
FM6	FM.factRankFinset_eq_card_of_indepFacts Ssot/FactMatroid.lean	GOL1	StaticLang.go_no_inheritance_hooks Ssot/LangStatic.lean
IND1	both_requirements_independent Ssot/Requirements.lean	IND2	both_requirements_independent' Ssot/Requirements.lean
JVA1	StaticLang.java_lacks_definition_hooks Ssot/LangStatic.lean	MFT3	MultiFact.exactRecovery_iff_properColoring Ssot/MultiFact.lean
MFT4	MultiFact.exists_exactRecovery_iff_colorable Ssot/MultiFact.lean	MFT5	MultiFact.binaryViews_nonclique_witness Ssot/MultiFact.lean
MFT6	MultiFact.binaryViews_colorable_two Ssot/MultiFact.lean	MFT7	MultiFact.binaryViews_not_colorable_one Ssot/MultiFact.lean
MFT8	MultiFact.binaryViews_oneTag_uniform_successProb_le_half Ssot/MultiFact.lean	MFT9	MultiFact.exists_exactOn_iff_colorableOn Ssot/MultiFact.lean
MFT10	MultiFact.binaryViews_oneTag_exactOn_card_le_two Ssot/MultiFact.lean	MFT11	MultiFact.pairProperColoring_of_componentColorings Ssot/MultiFact.lean
MFT12	MultiFact.pairColorable_of_colorable Ssot/MultiFact.lean	MFT13	MultiFact.pair_product_clique_budget_lower_bound Ssot/MultiFact.lean
MFT14	MultiFact.exists_exactOn_card_eq_maxColorableCard Ssot/MultiFact.lean	MFT15	MultiFact.binaryViews_maxColorableCard_one_eq_two Ssot/MultiFact.lean
MFT16	MultiFact.pair_product_success_card_lower_bound Ssot/MultiFact.lean	MFT17	MultiFact.pair_product_independent_lower_bound Ssot/MultiFact.lean
MFT18	MultiFact.exactOn_mass_le_maxColorableMass Ssot/MultiFact.lean	MFT19	MultiFact.exists_exactOn_mass_eq_maxColorableMass Ssot/MultiFact.lean
MFT20	MultiFact.colorable_iff_graphColorable Ssot/MultiFact.lean	MFT21	MultiFact.block_power_independent_lower_bound Ssot/MultiFact.lean
MFT22	MultiFact.block_add_independent_lower_bound Ssot/MultiFact.lean	MFT23	MultiFact.exactRateDistortionValue_upper Ssot/MultiFact.lean
MFT24	MultiFact.exists_exactRateDistortionValue Ssot/MultiFact.lean	MFT25	MultiFact.pairConfusable_iff_strongProdAdj Ssot/MultiFact.lean
MFT26	MultiFact.pairConfusabilityGraph_eq_strongProd Ssot/MultiFact.lean	MFT29	MultiFact.concatBlockConfusable_iff_strongProdAdj Ssot/MultiFact.lean
MFT30	MultiFact.blockConfusabilityGraph_addIsoStrongProd Ssot/MultiFact.lean	MFT31	MultiFact.blockMaxIndependentCard_eq_graphMaxIndependentCard Ssot/MultiFact.lean
MFT32	MultiFact.shannonLowerCapacity_eq_iSup_graphRates Ssot/MultiFact.lean	MFT33	MultiFact.blockRate_le_blockRate_mul Ssot/MultiFact.lean
MFT34	MultiFact.blockConfusable_iff_strongPowAdj Ssot/MultiFact.lean	MFT35	MultiFact.blockConfusabilityGraph_eq_strongPow Ssot/MultiFact.lean
MFT36	MultiFact.shannonLowerCapacity_eq_graphShannonLowerCapacity Ssot/MultiFact.lean	MFT38	MultiFact.strongPow_addIsoStrongProd Ssot/MultiFact.lean
MFT39	MultiFact.graphMaxIndependentCard_strongPow_add_eq_strongProd Ssot/MultiFact.lean	MFT43	MultiFact.graphNegLogSeq_subadditive Ssot/MultiFact.lean
MFT44	MultiFact.tendsto_graphPowerRateNat_graphShannonCapacityReal Ssot/MultiFact.lean	MFT45	MultiFact.graphPowerRate_le_graphShannonCapacityReal Ssot/MultiFact.lean
MFT46	MultiFact.graphShannonCapacityReal_nonneg Ssot/MultiFact.lean	MFT47	MultiFact.iSup_graphPowerRate_eq_graphShannonCapacityReal Ssot/MultiFact.lean

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ID	Lean Handle / Source	ID	Lean Handle / Source
MFT48	MultiFact.shannonCapacityReal_eq_iSup_blockRate Ssot/MultiFact.lean	MFT49	MultiFact.blockRate_le_shannonCapacityReal Ssot/MultiFact.lean
MFT50	MultiFact.compl_strongPow_adj_iff_exists Ssot/MultiFact.lean	MFT51	MultiFact.complStrongPow_colorable Ssot/MultiFact.lean
MFT52	MultiFact.graphMaxIndependentCard_strongPow_le_complChromaticPow Ssot/MultiFact.lean	MFT53	MultiFact.graphPowerRate_le_log_complChromatic Ssot/MultiFact.lean
MFT54	MultiFact.graphShannonCapacityReal_le_log_complChromatic Ssot/MultiFact.lean	MFT55	MultiFact.graphShannonCapacityReal_le_log_complChromaticNumber Ssot/MultiFact.lean
MFT56	MultiFact.graphShannonLowerCapacity_eq_ofReal_graphShannonCapacityReal Ssot/MultiFact.lean	MFT57	MultiFact.graphThetaPSDLogUpper_eq_graphThetaLogUpper Ssot/MultiFact.lean
MFT58	MultiFact.graphShannonCapacityReal_le_graphThetaPSDLogUpper Ssot/MultiFact.lean	MFT59	MultiFact.graphThetaPSDLogUpper_strongProd_le Ssot/MultiFact.lean
MFT60	MultiFact.graphThetaPSDLogSeq_subadditive Ssot/MultiFact.lean	MFT61	MultiFact.tendsto_graphThetaPSDPowerRateNat_graphThetaPSDAsymptoticUpper Ssot/MultiFact.lean
MFT62	MultiFact.graphThetaPSDAsymptoticUpper_le_graphThetaPSDLogUpper Ssot/MultiFact.lean	MFT63	MultiFact.iInf_graphThetaPSDPowerRate_eq_graphThetaPSDAsymptoticUpper Ssot/MultiFact.lean
MFT64	MultiFact.lovaszOrthoLogUpper_eq_graphThetaLogUpper Ssot/MultiFact.lean	MFT65	MultiFact.lovaszThetaPrimalLogUpper_eq_graphThetaPSDLogUpper Ssot/MultiFact.lean
MFT66	MultiFact.graphThetaLiftedDualLogUpper_eq_graphThetaSchurDualLogUpper Ssot/MultiFact.lean	MFT67	MultiFact.graphOneShotLogMaxIndependentCard_le_graphThetaLogUpper Ssot/MultiFact.lean
MFT68	MultiFact.graphPowerRate_le_graphThetaPSDPowerRate Ssot/MultiFact.lean	MFT69	MultiFact.graphShannonCapacityReal_le_graphThetaPSDAsymptoticUpper Ssot/MultiFact.lean
MFT70	MultiFact.lovaszThetaPrimalAsymptoticUpper_eq_graphThetaPSDAsymptoticUpper Ssot/MultiFact.lean	MFT71	MultiFact.graphShannonCapacityReal_le_lovaszThetaPrimalAsymptoticUpper Ssot/MultiFact.lean
MFT72	MultiFact.graphThetaSchurDualLogUpper_strongProd_le Ssot/MultiFact.lean	MFT73	MultiFact.graphThetaSchurDualLogUpper_iso_eq Ssot/MultiFact.lean
MFT74	MultiFact.graphThetaSchurDualLogSeq_subadditive Ssot/MultiFact.lean	MFT75	MultiFact.tendsto_graphThetaSchurDualPowerRateNat_graphThetaSchurDualAsymptoticUpper Ssot/MultiFact.lean
MFT76	MultiFact.iInf_graphThetaSchurDualPowerRate_eq_graphThetaSchurDualAsymptoticUpper Ssot/MultiFact.lean	MFT77	MultiFact.graphThetaSchurDualAsymptoticUpper_le_graphThetaPSDAsymptoticUpper Ssot/MultiFact.lean
MFT78	MultiFact.graphThetaSchurDualAsymptoticUpper_le_lovaszThetaPrimalAsymptoticUpper Ssot/MultiFact.lean	MFT79	MultiFact.lovaszThetaDualLogUpper_eq_graphThetaSchurDualLogUpper Ssot/MultiFact.lean
MFT80	MultiFact.lovaszThetaDualAsymptoticUpper_eq_graphThetaSchurDualAsymptoticUpper Ssot/MultiFact.lean	MFT81	MultiFact.lovaszThetaDualAsymptoticUpper_le_lovaszThetaPrimalAsymptoticUpper Ssot/MultiFact.lean
MFT82	MultiFact.lovaszThetaDualLogUpper_eq_lovaszThetaPrimalLogUpper Ssot/MultiFact.lean	MFT83	MultiFact.lovaszThetaDualAsymptoticUpper_eq_lovaszThetaPrimalAsymptoticUpper Ssot/MultiFact.lean
MFT84	MultiFact.graphShannonCapacityReal_le_lovaszThetaAsymptoticUpper Ssot/MultiFact.lean	MFT85	MultiFact.graphShannonCapacityReal_eq_lovaszThetaAsymptoticUpper_clusterGraph Ssot/MultiFact.lean
MFT86	MultiFact.graphShannonCapacityReal_eq_log_card_clusterGraph Ssot/MultiFact.lean	MFT87	MultiFact.lovaszThetaAsymptoticUpper_eq_log_card_clusterGraph Ssot/MultiFact.lean

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ID	Lean Handle / Source	ID	Lean Handle / Source
MFT88	MultiFact.confusabilityGraph_eq_clusterGraph_transcriptLabel Ssot/MultiFact.lean	MFT89	MultiFact.shannonCapacityReal_eq_shannonLovaszThetaAsymptoticUpper_of_fiberCoherent Ssot/MultiFact.lean
MFT90	MultiFact.shannonCapacityReal_eq_log_transcriptFiberCard_of_fiberCoherent Ssot/MultiFact.lean	MFT91	MultiFact.shannonLovaszThetaAsymptoticUpper_eq_log_transcriptFiberCard_of_fiberCoherent Ssot/MultiFact.lean
MFT96	MultiFact.shannonLovaszThetaAsymptoticUpper_eq_log_connectedComponents_of_confusableTransitive Ssot/MultiFact.lean	MFT100	MultiFact.shannonCapacityReal_eq_shannonLovaszThetaAsymptoticUpper_of_meetWitnessed Ssot/MultiFact.lean
MFT101	MultiFact.shannonCapacityReal_eq_log_connectedComponents_of_meetWitnessed Ssot/MultiFact.lean	MFT102	MultiFact.shannonLovaszThetaAsymptoticUpper_eq_log_connectedComponents_of_meetWitnessed Ssot/MultiFact.lean
MFT107	MultiFact.ViewCompositionClosed Ssot/MultiFact.lean	MFT108	MultiFact.confusableTransitive_iff_viewCompositionClosed Ssot/MultiFact.lean
MFT109	MultiFact.confusable_iff_exists_viewClusterAdj Ssot/MultiFact.lean	MFT110	MultiFact.SupportConfusable Ssot/MultiFact.lean
MFT111	MultiFact.deterministicSupportConfusable_iff_confusable Ssot/MultiFact.lean	MFT112	MultiFact.deterministicSupportConfusabilityGraph_eq_confusabilityGraph Ssot/MultiFact.lean
MFT113	MultiFact.card_state_eq Ssot/MultiFact.lean	MFT114	MultiFact.confusableOracle_eq_true_iff Ssot/MultiFact.lean
MFT115	MultiFact.confusableOrderedPairs Ssot/MultiFact.lean	MFT116	MultiFact.card_state_prod_eq Ssot/MultiFact.lean
MFT117	MultiFact.card_confusableOrderedPairs_le_naive_bound Ssot/MultiFact.lean	MFT118	MultiFact.observe_eq_iff_subset_agreeSet Ssot/MultiFact.lean
MFT119	MultiFact.confusable_iff_exists_view_subset_agreeSet Ssot/MultiFact.lean	MFT120	MultiFact.agreeSet_statePerm_eq Ssot/MultiFact.lean
MFT121	MultiFact.confusable_statePerm_iff Ssot/MultiFact.lean	MFT122	MultiFact.confusable_congr_agreeSet Ssot/MultiFact.lean
MFT123	MultiFact.confusabilityGraph_statePermIso Ssot/MultiFact.lean	MFT124	MultiFact.exists_confusabilityGraph_iso_send Ssot/MultiFact.lean
MFT125	MultiFact.agreementUpwardFamily_upwardClosed Ssot/MultiFact.lean	MFT126	MultiFact.confusable_iff_agreeSet_mem_agreementUpwardFamily Ssot/MultiFact.lean
MFT127	MultiFact.confusable_viewFamilyOfAgreementUpwardFamily_iff Ssot/MultiFact.lean	MFT128	MultiFact.exists_viewFamily_iff_agreementClassified Ssot/MultiFact.lean
MFT129	MultiFact.agreeSet_eq_univ_iff Ssot/MultiFact.lean	MFT130	MultiFact.exists_distinct_states_with_agreeSet_eq_iff Ssot/MultiFact.lean
MFT131	MultiFact.upwardClosed_family_eq_on_proper_subsets_of_same_relation Ssot/MultiFact.lean	MFT132	MultiFact.agreementOracle_eq_true_iff Ssot/MultiFact.lean
MFT133	MultiFact.agreementOracle_eq_confusableOracle Ssot/MultiFact.lean	MFT134	MultiFact.agreementOracleViewWork_le Ssot/MultiFact.lean
MFT135	MultiFact.agreementOracleTotalWork_le Ssot/MultiFact.lean	MFT136	MultiFact.meetWitnessed_implies_transitive_confusability Ssot/MultiFact.lean
MFT137	MultiFact.transitive_confusability_implies_cluster_graph Ssot/MultiFact.lean	NPM1	Deps.npm_ssot_incomplete Ssot/LangDeps.lean
NPM2	Deps.npm_has_introspection Ssot/LangDeps.lean	NPM3	Deps.npm_no_hooks Ssot/LangDeps.lean
OBS1	ObserverModel.exact_recovery_implies_pair_injective paper1/Paper1IT/ObserverTagModel.lean	OBS2	ObserverModel.pair_injective_implies_exact_recovery paper1/Paper1IT/ObserverTagModel.lean

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ID	Lean Handle / Source	ID	Lean Handle / Source
OBS3	ObserverModel.fiber_card_le_tag_alphabet paper1/Paper1IT/ObserverTagModel.lean	OBS4	ObserverModel.exact_recovery_global_count paper1/Paper1IT/ObserverTagModel.lean
OBS5	ObserverModel.clique_card_le_tag_alphabet paper1/Paper1IT/ObserverTagModel.lean	PNM1	Deps.pnpm_ssot_incomplete Ssot/LangDeps.lean
PNM2	Deps.pnpm_has_introspection Ssot/LangDeps.lean	PNM3	Deps.pnpm_no_hooks Ssot/LangDeps.lean
POT1	Deps.poetry_ssot_incomplete Ssot/LangDeps.lean	POT2	Deps.poetry_has_introspection Ssot/LangDeps.lean
POT3	Deps.poetry_no_hooks Ssot/LangDeps.lean	PRB47	ObserverModel.sourceEntropy_ eq_observationTagEntropy_add_ conditionalEntropyGivenPair paper1/Paper1IT/ProbabilisticFinite.lean
PRB55	ObserverModel.successSet_clique_entropy_le_ log_tags paper1/Paper1IT/FanoFinite.lean	PRB68	ObserverModel.deterministicKernel_entropy_ data_processing paper1/Paper1IT/ProbabilisticFinite.lean
PRB73	ObserverModel.observationTagEntropy_gap_nonneg paper1/Paper1IT/ProbabilisticFinite.lean	PRB74	ObserverModel.observationTagEntropy_gap_eq_ zero_of_klDiv_zero_uniform paper1/Paper1IT/ProbabilisticFinite.lean
PRB81	ObserverModel.decodedOutputEntropy_le_ mutualInfoDeterministic paper1/Paper1IT/ProbabilisticFinite.lean	PRB82	ObserverModel.decodedOutputEntropy_source_gap_ nonneg paper1/Paper1IT/ProbabilisticFinite.lean
PRB83	ObserverModel.decodedOutputEntropy_log_gap_ nonneg paper1/Paper1IT/ProbabilisticFinite.lean	PRB85	ObserverModel.decodedOutputEntropy_fano_ budgeted paper1/Paper1IT/FanoFinite.lean
PRB87	ObserverModel.decodedOutputEntropy_fano_ observation_only paper1/Paper1IT/FanoFinite.lean	PRB90	ObserverModel.DeterministicObservable.entropy_ coarsen_le_sourceEntropy paper1/Paper1IT/ProbabilisticFinite.lean
PRB93	ObserverModel.decodedOutputEntropy_gap_eq_ zero_of_klDiv_zero_uniform paper1/Paper1IT/ProbabilisticFinite.lean	PRB94	ObserverModel.decodedOutputEntropy_gap_pos_ implies_klDiv_ne_zero paper1/Paper1IT/ProbabilisticFinite.lean
PRB95	ObserverModel.decodedOutputObservable_entropy_ le_sourceEntropy paper1/Paper1IT/ProbabilisticFinite.lean	PYH1	Python.python_has_hooks Ssot/LangPython.lean
PYI1	Python.python_has_introspection Ssot/LangPython.lean	RAT1	ClaimClosure.rate_incoherence_step Ssot/ClaimClosure.lean
RED1	ClaimClosure.redundancy_incoherence_equiv Ssot/ClaimClosure.lean	REQ1	structural_facts_fixed_at_definition Ssot/Foundations.lean
REQ2	definition_hooks_necessary Ssot/Requirements.lean	REQ3	introspection_necessary_for_verification Ssot/Requirements.lean
RUH1	Rust.rust_has_definition_hooks Ssot/LangRust.lean	RUI1	Rust.rust_lacks_introspection Ssot/LangRust.lean
SOT1	ssot_iff Ssot/Completeness.lean	TRI1	timing_trichotomy_exhaustive Ssot/Foundations.lean
TSE1	StaticLang.ts_types_erased Ssot/LangStatic.lean	TSN1	StaticLang.ts_no_runtime_type_definitions Ssot/LangStatic.lean

2 Claim Mapping to Lean Handles

This section provides the direct mapping between paper claims and the mechanized proof handles in the Lean 4 artifact.

Paper claim	Lean handle
Proposition 8.3: View-fiber clique sizes	FM6, AFM7, AFM8, AFM11, AFM9, AFM6, AFM12, AFM10
Corollary 8.5: Rank additivity	AFM17
Confusability Clique Bound	OBS5
Information-Constrained DOF Lower Bound	CIA4
Proposition 3.4: Equivalent formulations	CIA3, PRB90, OBS5, PRB93, PRB94, PRB81, PRB83, PRB82, PRB95, PRB68, OBS4, OBS1, OBS3, PRB74, PRB73, PRB47, PRB55
Proposition 8.4: Matroid direct sum under block composition	AFM19, AFM18
Proposition 8.1: Affine fact-matroid specialization	AFM3, AFM4, AFM1, AFM2, AFM5
Definition 5.2: Shannon Capacity	MFT49, MFT43, MFT45, MFT46, MFT56, MFT47, MFT48, MFT44
Zero-Delay State Synchronization	REQ2
Definition 2.7: Independent Location	RAT1, CAP3, CAP2
Theorem 3.2: Confusability formulation	OBS5
Derivation	DER2
Corollary 3.3: Counting converse	CIA3
Proposition 3.5: Budgeted finite-error converse	PRB85, PRB87
Requirements are Independent	IND1, IND2
Above-Threshold Lower Bound	BND2
Definition 4.8: Coordinate-Subset View	MFT20, MFT3, MFT4
Theorem 4.10: Exact finite-error success budget	MFT15, MFT14
Theorem 4.11: First non-clique witness	MFT6, MFT5, MFT7, MFT10, MFT8
Definition 4.13: Strong Product	MFT12, MFT11, MFT13, MFT16
Theorem 4.9: Success-set characterization	MFT9
Theorem 3.1: Pair-injectivity characterization	OBS1, OBS2
Structural Side-Information	REQ3
Definition 11.2: Verifiable Structural Integrity	SOT1
Structural Fact	REQ1, TRI1
Unbounded Gap	BND3
Modification Cost Model	BND1

Auto summary: mapped 27/27 (full=27, derived=0, unmapped=0).